

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 19/05/2026
Team ID: HDI Predictor Team
Project: HDI Predictor using Machine Learning and Flask

Component	Technology
Frontend	HTML, CSS, Bootstrap, JavaScript
Backend	Python, Flask
Machine Learning	Scikit-learn, Pandas, NumPy
Visualization	Matplotlib, Seaborn
Database/File	CSV Dataset, Pickle Model
Deployment	Local Flask Server, GitHub

Application Characteristics

Characteristic	Implementation
Open Source	Flask, Scikit-learn, Pandas, NumPy
Security	Input validation, Flask rate limiting
Scalability	Modular Flask services
Availability	Local deployment with reusable model
Performance	Pre-trained pickle model for fast inference